

WEATHERFORD STAFFORD, OK, USA

WMO: 722368

Lat: 35.550N

Lon: 98.667W

Elev: 1605

StdP: 13.86

Time Zone: -6.00 (NAC)

Period: 08-19

WBAN: 00364

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.1	18.4	1.4	6.3	15.6	7.0	8.4	21.3	28.2	41.0	26.3	42.2	9.1	0	0.540

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.8	100.3	73.3	98.9	73.2	95.5	73.2	77.0	91.1	75.9	90.2	75.1	89.3	11.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.1	130.4	81.2	72.4	127.3	80.5	71.9	125.4	80.0	41.5	91.1	40.5	91.3	39.7	89.5	87.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
25.7	23.2	20.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	60.2	37.9	41.3	50.6	58.6	68.4	79.8	82.5	80.9	73.6	60.2	48.6	39.2	
	DBStd	18.22	9.92	11.58	10.50	8.95	8.49	5.49	5.81	6.11	7.09	9.67	9.12	8.86	
	HDD50	1322	386	286	119	27	1	0	0	0	0	25	129	349	
	HDD65	3767	839	664	455	227	65	0	1	0	13	207	495	801	
	CDD50	5060	12	43	137	285	572	894	1009	959	708	342	87	12	
	CDD65	2030	0	1	9	35	171	444	544	494	272	58	2	0	
	CDH74	23430	1	23	107	407	1725	5206	6863	5788	2721	561	28	0	
	CDH80	11711	0	4	21	95	673	2634	3832	3114	1190	145	3	0	
Wind	WSAvg	9.2	8.9	9.6	10.8	11.5	10.1	10.4	8.1	7.5	7.9	8.9	9.2	8.3	
Precipitation	PrecAvg	27.2	0.9	1.1	2.0	2.1	4.3	3.8	2.0	3.0	3.1	2.9	1.6	1.1	
	PrecMax	42.2	3.1	4.2	6.8	8.6	16.5	11.2	6.3	7.4	9.2	9.1	6.9	3.7	
	PrecMin	14.8	0.0	0.0	0.0	0.2	0.3	0.4	0.0	0.3	0.0	0.1	0.0	0.0	
	PrecStd	6.8	0.9	1.0	1.6	1.6	3.3	2.3	1.6	1.9	2.4	2.6	1.6	1.0	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.7	77.3	82.3	87.9	95.0	102.0	105.7	105.5	99.0	88.5	78.8	67.9	
		MCWB	51.5	56.1	61.3	64.0	70.5	73.1	73.5	73.5	71.5	67.8	60.7	52.3	
	2%	DB	63.7	70.4	77.0	82.0	90.5	98.6	100.3	100.3	94.6	84.0	71.9	63.0	
		MCWB	48.1	54.8	58.9	63.0	69.8	72.6	73.3	73.7	72.4	67.4	58.1	52.5	
	5%	DB	59.1	64.1	72.2	79.0	88.0	95.3	98.8	98.8	90.4	80.8	67.7	58.5	
		MCWB	46.6	52.4	57.6	61.8	69.1	73.2	73.3	73.7	71.0	65.1	56.5	48.0	
	10%	DB	54.2	60.5	68.0	74.5	82.4	91.4	95.4	94.9	87.8	76.9	63.3	54.0	
		MCWB	43.9	50.4	55.2	60.7	67.6	72.3	73.6	73.1	70.5	63.0	54.3	46.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.1	61.2	65.7	69.7	75.3	78.3	79.1	77.7	76.3	73.3	66.1	61.1
			MCDB	64.8	64.9	77.8	78.7	87.1	91.9	90.8	92.2	89.1	81.6	71.1	63.2
		2%	WB	50.1	57.9	63.1	67.4	73.8	76.5	77.2	76.3	74.9	71.7	61.3	54.4
			MCDB	61.4	66.8	71.6	77.1	84.2	90.3	91.2	91.3	86.4	79.0	64.6	58.4
5%		WB	47.4	54.9	60.0	65.6	72.2	75.3	76.0	75.3	73.7	69.2	58.7	49.8	
		MCDB	58.2	62.7	67.3	74.9	82.1	88.9	90.3	90.8	84.0	75.8	65.1	56.7	
10%		WB	44.8	50.7	57.1	63.2	70.4	74.3	75.0	74.4	72.4	66.3	55.5	46.1	
		MCDB	53.0	60.8	66.5	70.7	79.5	87.9	89.6	89.3	82.4	73.9	62.0	52.6	
Mean Daily Temperature Range	5% DB	MDBR	21.4	21.1	23.1	22.9	22.3	21.7	21.8	21.9	21.3	22.3	21.4	18.9	
		MCDBR	29.9	29.1	27.6	28.9	26.0	24.5	25.0	25.7	24.8	26.2	26.2	26.4	
		MCWBR	17.7	16.9	15.6	14.8	10.9	7.7	6.7	7.2	8.5	13.2	14.8	15.8	
	5% WB	MCDBR	28.3	25.7	23.4	23.6	22.3	21.5	21.6	22.5	19.6	20.2	21.5	22.6	
		MCWBR	17.5	17.1	15.4	13.6	11.2	8.5	7.5	7.6	7.7	11.5	15.0	16.1	
Clear-Sky Solar Irradiance	taub	0.282	0.291	0.332	0.368	0.395	0.416	0.414	0.415	0.377	0.347	0.306	0.283		
	taud	2.523	2.508	2.409	2.337	2.334	2.317	2.343	2.292	2.374	2.443	2.496	2.518		
	Ebn at Noon	295	304	298	290	282	275	275	273	279	279	283	287		
	Edh at Noon	25	29	35	40	40	41	40	41	36	30	26	24		
All-Sky Solar Radiation	RadAvg	921	1118	1479	1827	1985	2214	2212	1999	1673	1262	964	778		
	RadStd	87	155	102	140	204	184	122	119	116	166	125	81		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (35 neighbors)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page